

GENERAL THETA FOUNDATIONS I: A BOUNDEDNESS CRITERION FOR NUMERICAL REALIZATION

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ABSTRACT. A prescribed command length can make an infinite orthogonal action require only bounded stochastic memory: deterministic motion may be placed in the horizon-dependent decoder. We identify the remaining obstruction. On the invariant span of the seeds, let the synchronization subgroup be the closed normal subgroup generated by the multiplicative differences $U_a^{-1}U_b$ of command matrices. Exact numerical response probabilities admit uniformly bounded clocked positive-realization width if and only if this subgroup is finite. Otherwise the minimum width tends to infinity, also at every fixed error below an explicit seed-activation threshold. The converse permits arbitrary hidden states, time-dependent rows, and redesign at each horizon; it assumes neither commutativity nor a spectral gap. The criterion is decidable for matrices and seeds in a specified real number field, by exact finite-group enumeration with a proved stopping bound. We also give the exact finite-horizon continuation-polytope formulation and a comparison between optimized packet loss and common enclosure dilation. The previous metric Diophantine and Liouville theorems are retained with full proofs. The resource throughout is nonuniform clocked atomic-row label width, not table size or uniform computational space.

1. INTRODUCTION

For a nonempty finite alphabet $\mathcal{A}$ with orthogonal command matrices U_a , a numerical experiment asks for one binary response after a word $w = a_1 \cdots a_N$:

$$(1.1) \quad \mathbb{P}(B = b \mid x, w, j) = \frac{1 + b\rho\langle e_j, U_w x \rangle}{2}, \quad U_w = U_{a_N} \cdots U_{a_1}, \quad b \in \{-1, 1\}.$$

Here $X \subset \mathbb{S}^{D-1}$ is a nonempty finite seed set and $0 < \rho \leq 1$ is fixed. A simulator receives the seed, reads every command once, and finally receives one coordinate query j . We count its largest persistent label alphabet, including a held binary answer. The simulator may use arbitrary stochastic rows, different rows at different epochs, and a different design for every prescribed horizon. Section 2 gives the precise convention.

The distinction between a signed linear representation and a positive stochastic realization is classical. The mean in (1.1) has signed dimension at most D . Positivity and common command updates impose different constraints. Equally importantly, a free external clock changes the relevant group. A single irrational rotation generates an infinite compact group, but its word of length N is predetermined. Its entire effect can be absorbed in the decoder. The generated group alone cannot characterize bounded clocked width.

Let R be the smallest invariant real subspace containing X , and restrict all commands to R . Put

$$H = \overline{\langle U_a|_R : a \in \mathcal{A} \rangle}.$$

The closure is a compact matrix group. Define its *synchronization subgroup*

$$(1.2) \quad \mathcal{L} = \overline{\langle h(U_a^{-1}U_b)h^{-1} : a, b \in \mathcal{A}, h \in H \rangle}.$$

It is the least closed normal subgroup for which all commands have the same image in $H/\mathcal{L}$. This group removes the common motion of words having the same length. It is defined on the active space, not on unused ambient coordinates.

Theorem 1.1 (Boundedness dichotomy). *For the experiment (1.1), the following are equivalent:*

- (1) $\mathcal{L}$ is finite;
- (2) $\sup_{N \geq 1} W_{N,0} < \infty$.

If $\mathcal{L}$ is finite, deterministic command updates give

$$W_{N,0} \leq \max\{2, |X| |\mathcal{L}|\} \quad (N \geq 1).$$

If $\mathcal{L}$ is infinite, let $E = R \ominus R^{\mathcal{L}^\circ}$ and $\eta_X = \max_{x \in X} \|P_E x\|$. Then $\eta_X > 0$ and

$$(1.3) \quad W_{N,\varepsilon} \longrightarrow \infty \quad \text{for every fixed } 0 \leq \varepsilon < \frac{\rho \eta_X}{2\sqrt{D}}.$$

No spectral gap, connectedness of H , or arithmetic hypothesis is required.

For symmetric coordinate seeds $X = \{\pm e_1, \dots, \pm e_D\}$, the sufficient threshold in the infinite case is uniformly $\varepsilon < \rho/(2D)$. Theorem 5.2 makes the exact boundedness dichotomy decidable when all input data lie in a specified real number field. It does not decide positivity of an infinite-dimensional action gap, nor determine the growth rate in every infinite case.

The proof identifies a finite-word obstruction before introducing probability. After removing a reference command, the sets of reachable vectors at equal lengths are nested. If their cardinalities remain bounded, their stabilized finite set is invariant under the synchronization subgroup. On the complement of its connected-component fixed space this is impossible. Compactness therefore gives a single finite packet whose every orbit has more than any proposed number of endpoint labels. The inherited endpoint-centroid contraction then forces a positive loss at each independent packet. This turns a finite-orbit criterion into a converse against arbitrary hidden stochastic states.

Two complementary realization statements are retained and made explicit from the concurrently published revision 44 [22]. Theorem 6.1 gives the exact finite-horizon continuation-polytope formula for a prescribed width profile, including its actual terminal approximation error. Its ambient coordinates are all suffix responses; restricting them to the small signed representation would lose hidden realizations. Theorem 7.1 proves

$$\sup_{B \geq 1} \frac{-\log(1 - \Gamma_A(k, B))}{B} \leq \log \lambda_A(k, a),$$

linking the previously separate packet and enclosure profiles. Equality is not asserted. The right side constructs vector-state realizations, whereas the continuation formula covers every hidden realization.

Provenance of the statements. The endpoint contraction, finite interval budget, circle optimization, and arithmetic results are retained from the predecessor development [7, 8, 21], with full proofs below. The finite-horizon factorization theorem is a normalized controlled version of classical shift-cone realization, not a claim to have discovered positive realization [12, 13, 20]. The finite-bit compiler of revision 44 is also retained with its complete proof and code, and its scratch space remains a separate charged resource. The new organizing result is Theorem 1.1, including removal of common clock motion, its arbitrary-hidden-state converse, and its finite algebraic decision consequence. The congruence bound used in that decision is classical and is proved explicitly. The packing-dilation balance of Corollary 3.7 remains a conditional corollary, not an exact quantitative classification of all alphabets. Section 8 compares the optimization objects and approximation norms before the retained arithmetic applications.

2. THE NUMERICAL EXPERIMENT AND THE REGISTER

Let $V = \mathbb{R}^D$, let $\mathcal{A}$ be a nonempty finite alphabet, and fix orthogonal matrices U_a , $a \in \mathcal{A}$. Write $U_w = U_{a_N} \cdots U_{a_1}$ in the order of execution. Let $X \subset \mathbb{S}^{D-1}$ be a nonempty finite seed set and $0 < \rho \leq 1$. The required law is (1.1). Commands are externally supplied, not chosen by a policy inside the simulator. No distribution on command words is part of the specification.

Definition 2.1 (Clocked atomic-row realization). At horizon N , a machine consists of finite sets S_t with $|S_t| = K_t$ for $0 \leq t \leq N$, an initialization probability row E_x on S_0 , stochastic matrices $T_{t,a} : S_{t-1} \rightarrow S_t$, and terminal columns $d_j : S_N \rightarrow [-1, 1]$. Its numerical mean is

$$E_x T_{1,a_1} \cdots T_{N,a_N} d_j.$$

After the atomic query j , it draws and holds the binary answer with that mean. The final answer cut has two labels. Its peak is $\max(2, K_0, \dots, K_N)$. Available labels count even when unreachable; adding unused normalized rows is permitted.

All persistent randomness, input history and protocol distinctions must be in the current label. Only the external epoch, the prescribed horizon and the read-only row tables are free. Table construction, lookup, arithmetic and exact sampling of known real stochastic rows are uncharged. Matrices, rows and state sets may depend on N ; this is nonuniform positive-realization width, not uniform computational space. A common-row construction uses the same command matrices at all command epochs but may still have an N -dependent decoder. It need not answer at earlier stopping lengths.

All logarithms in finite budgets are natural. In particular, every width is at least two because the held answer is charged.

Let $W_{N,\varepsilon}$ be the least peak among machines whose every binary row differs from (1.1) by at most ε in total variation. For fixed finite $K_\bullet$ the optimal actual error $\mathcal{E}_N(K_\bullet)$ is attained: the row parameters lie in a finite product of compact sets and the error is a finite maximum of continuous functions. For planar results we use $X = \{\pm e_1, \pm e_2\}$ and include the identity letter. We keep $\rho \leq 1/10$ and $\varepsilon < \rho/(2\sqrt{2})$.

Lemma 2.2 (Calibration under any external test law). *Fix a seed $x \in X$ and any law on the complete command word, independent of private initialization. Let $Y_t = U_{a_t} \cdots U_{a_1} x$, $Z_t = \mathbb{E}[Y_t | S_t]$ and $q_t = \mathbb{E}\|Z_t\|$. Every machine with row error at most ε satisfies*

$$(2.1) \quad q_N \geq \kappa := \rho/\sqrt{D} - 2\varepsilon.$$

At a boundary preceding an independent complete packet, the next boundary admits the conditional-mean identity used in Lemma 3.2. No single-step independence inside a packet is asserted.

Proof. For a fixed word, every coordinate of $\mathbb{E}[d(S_N) | w] - \rho Y_N$ has modulus at most 2ε . Since $\|Y_N\| = 1$, averaging gives

$$\mathbb{E}\langle Y_N, d(S_N) \rangle \geq \rho - 2\varepsilon\sqrt{D}.$$

On the other hand this pairing equals $\mathbb{E}\langle Z_N, d(S_N) \rangle$ and is at most $\sqrt{D}q_N$. This proves (2.1). Combining terminal columns is a proof device, not a requirement to provide their joint output distribution. $\square$

3. FINITE WORD PROFILES AND COMPATIBLE ENCLOSURES

The results in this section apply to every finite orthogonal alphabet, including noncommuting alphabets. No connectedness, arithmetic condition or spectral gap is assumed. Invariant directions may make a converse profile zero; this is a property of the stated profile, not an omitted hypothesis.

3.1. An optimized distortion. For $B, k \geq 1$ and a probability vector p on the actual words $\mathcal{A}^B$, define

$$(3.1) \quad D_k(p) = 1 - \max_{\substack{u \in \mathbb{S}^{D-1} \\ v_1, \dots, v_k \in \mathbb{S}^{D-1}}} \sum_{w \in \mathcal{A}^B} p_w \max_{1 \leq j \leq k} \langle v_j, U_w u \rangle,$$

$$(3.2) \quad \Gamma_{\mathcal{A}}(k, B) = \max_{p \in \Delta(\mathcal{A}^B)} D_k(p), \quad \chi_{\mathcal{A}}(k, B) = -\log(1 - \Gamma_{\mathcal{A}}(k, B)).$$

We allow $\chi = +\infty$. Repeated product matrices can be merged for the optimization: their probability masses add. Their representative words must still be executed one ordinary command at a time.

Proposition 3.1 (Finite-data properties). *The maxima in (3.1)–(3.2) are attained, $0 \leq \Gamma_{\mathcal{A}}(k, B) \leq 1$, and the profile is invariant under simultaneous orthogonal changes of coordinates. It is nonincreasing in k and unchanged by duplicating a letter. For two indexed orthogonal alphabets U, V ,*

$$(3.3) \quad |\Gamma_U(k, B) - \Gamma_V(k, B)| \leq B \max_a \|U_a - V_a\|_{\text{op}}.$$

For a fully listed finite alphabet of real-algebraic matrices and fixed integers k, B, D , $\Gamma_{\mathcal{A}}(k, B)$ and an optimizing word law are real-algebraic and can be determined by real quantifier elimination.

Proof. All parameter spaces are compact and the objective is continuous. Its maximum over directions is convex in p , so D_k is continuous and concave. The maximum scalar product is at most one. By choosing a single direction parallel to $\sum p_w U_w u$, the inner maximum is nonnegative; hence $D_k \in [0, 1]$. Conjugating all directions proves invariance. Additional centers only increase that inner maximum. Product telescoping gives $\|U_w - V_w\|_{\text{op}} \leq B \max_a \|U_a - V_a\|_{\text{op}}$, and maxima and minima preserve this uniform bound.

For the last assertion, enumerate the product matrices. Unit-sphere equations, simplex inequalities and finitely many maximum operations define a compact semialgebraic optimization with algebraic coefficients. Maximum operations can be expressed by finite alternatives or auxiliary variables. Quantifier elimination over the real closed field of real-algebraic numbers determines the endpoint of the feasible value interval and a witness [1]. The word table may have $|\mathcal{A}|^B$ entries; this is not a polynomial-time assertion or a decision procedure for an infinite spectral gap. No logarithm needs to be eliminated: finite budget comparisons can instead use products of $1 - \Gamma$. $\square$

Lemma 3.2 (Endpoint contraction; inherited mechanism). *Let a complete random word W with law p be independent of all information before its start. Internal letters of W may be correlated. If the endpoint register has at most k labels, then*

$$(3.4) \quad q' \leq (1 - D_k(p))q.$$

There is no bound on the intermediate registers.

Proof. Write $z_s = \mathbb{E}[Y \mid S = s]$ at the start. Composing all internal stochastic rows yields a common endpoint kernel $T(j \mid s, w)$. Since the entire word and the machine's fresh randomness are independent of the past,

$$z'_j = \mathbb{E}[U_W z_S \mid S' = j].$$

This is an endpoint identity; it would not generally hold after conditioning on a correlated next letter inside the packet. Choose $v_j = z'_j / \|z'_j\|$ for nonzero z'_j , and any unit vector otherwise. Then

$$\begin{aligned} q' &= \sum_s \mathbb{P}(S = s) \sum_w p_w \sum_j T(j \mid s, w) \langle v_j, U_w z_s \rangle \\ &\leq \sum_s \mathbb{P}(S = s) \|z_s\| \max_{\|u\|=1} \sum_w p_w \max_j \langle v_j, U_w u \rangle \leq (1 - D_k(p))q. \end{aligned}$$

Terms with $z_s = 0$ contribute zero. Replacing an executable assignment by the largest pairing is a relaxation in the correct direction. Arbitrary unobservable hidden coordinates are included in s ; no statewise observable-vector hypothesis is used. This is the conditional-centroid mechanism of [7, 8], now applied to the optimized finite word datum. $\square$

For an available profile $K_\bullet$, let $I_0 = 0$ and define

$$(3.5) \quad I_t = \max \left\{ I_{t-1}, \max_{1 \leq B \leq t} [I_{t-B} + \chi_{\mathcal{A}}(K_t, B)] \right\}, \quad \mathcal{I}_N(K_\bullet) = I_N.$$

This is the largest sum of charges of disjoint command intervals, with an interval of length B ending at t charged by $\chi_{\mathcal{A}}(K_t, B)$. The recurrence depends only on the alphabet and the available profile.

If $\Gamma_{\mathcal{A}}(k, B) = 1$, a positive calibrated terminal correlation is impossible after one such packet; the corresponding cost is $+\infty$.

Theorem 3.3 (A nonuniform interval budget). *Every machine with $\kappa = \rho/\sqrt{D} - 2\varepsilon > 0$ satisfies*

$$(3.6) \quad \mathcal{I}_N(K_\bullet) \leq \log(1/\kappa).$$

More generally its optimal actual terminal error obeys

$$(3.7) \quad \mathcal{E}_N(K_\bullet) \geq \frac{1}{2} \left[\frac{\rho}{\sqrt{D}} - e^{-\mathcal{I}_N(K_\bullet)} \right]_+.$$

For a uniform peak bound K , every $B \leq N$ gives

$$(3.8) \quad \lfloor N/B \rfloor \chi_{\mathcal{A}}(K, B) \leq \log(1/\kappa).$$

For a fixed B, k with $\Gamma_{\mathcal{A}}(k, B) > 0$,

$$(3.9) \quad \#\{1 \leq t \leq N : K_t \leq k\} \leq B - 1 + \frac{B \log(1/\kappa)}{\chi_{\mathcal{A}}(k, B)}.$$

If $\chi = +\infty$, the last fraction is zero.

Proof. Choose any maximizing disjoint interval family in (3.5). On each selected interval put an independently sampled optimizing complete word from Proposition 3.1. All gaps and the final remainder contain fixed deterministic command words. A deterministic orthogonal command followed by a stochastic update cannot increase the conditional-mean norm, by conditional Jensen. Lemma 3.2 therefore gives

$$q_N \leq \prod_{(t-B, t]} (1 - \Gamma_{\mathcal{A}}(K_t, B)) = e^{-\mathcal{I}_N(K_\bullet)},$$

since $q_0 = 1$. The chosen law can depend on the available layer sizes, not on private states. Wordwise correctness requires the same machine to pass it. Lemma 2.2 proves (3.6) and (3.7); taking consecutive intervals proves (3.8).

For (3.9), discard candidate endpoints before B . Select the earliest remaining endpoint, delete all later candidates less than B away, and repeat. If M is the original number and J the number selected, the explicit count is

$$M - (B - 1) \leq BJ.$$

The associated length- B intervals are disjoint and each contributes at least $\chi_{\mathcal{A}}(k, B)$. Applying (3.6) gives the result. No restriction on the other registers was introduced. $\square$

The cap union estimate below is the geometric certificate already used in the predecessor packet method [7]. The finite incidence formulation specifies how to optimize that certificate on a fully listed word set.

Proposition 3.4 (Uniform cap certificates). *Suppose a law p on length- B words satisfies*

$$\sup_{u,v \in \mathbb{S}^{D-1}} p\{w : \|U_w u - v\| < \delta\} \leq b.$$

Then $\Gamma_{\mathcal{A}}(k, B) \geq \frac{1}{2}\delta^2(1 - kb)_+$. For algebraic product matrices and algebraic δ , the best such b is the value of a finite linear program once cap incidence types have been decided by real quantifier elimination.

Proof. The union of k open caps has probability at most kb . Outside it, $1 - \max_j \langle v_j, U_w u \rangle = \frac{1}{2} \min_j \|U_w u - v_j\|^2 \geq \delta^2/2$. For the algorithmic assertion, a subset of the finite word list either can or cannot be contained in one such cap for some u, v . This is a finite system of algebraic inequalities, including the sphere equations. For every feasible subset A impose $\sum_{w \in A} p_w \leq b$, along with the simplex constraints, and minimize b . Every actual incidence set is feasible; conversely a feasible subset is included in an actual incidence set and has no greater mass. Thus these constraints give exactly the desired supremum. The exponentially large enumeration is not concealed. $\square$

3.2. A separate exact-synthesis profile. Fix $0 < a < 1$. For k such that the following family is nonempty, let

$$(3.10) \quad \lambda_{\mathcal{A}}(k, a) = \min_P \inf \{\lambda \geq 1 : U_b P \subseteq \lambda P \text{ for all } b \in \mathcal{A}\},$$

where P ranges over convex hulls of at most k points in $\mathbb{B}^D$ containing $a\mathbb{B}^D$. An empty family has value $+\infty$.

Theorem 3.5 (Compatible enclosure synthesis). *The finite value in (3.10) is attained and lies in $[1, a^{-1}]$. If $\rho\lambda_{\mathcal{A}}(k, a)^N \leq a$, there is an exact machine with $\max(k, 2)$ labels, common command matrices and a horizon-dependent decoder. Each row has at most $D + 1$ successors. Algebraic matrices and algebraic a give an algebraically decidable profile value and algebraic row witnesses. For two indexed alphabets with a common feasible family,*

$$(3.11) \quad |\lambda_U(k, a) - \lambda_V(k, a)| \leq a^{-1} \max_b \|U_b - V_b\|_{\text{op}}.$$

Proof. Parameterize P by k points in the closed unit ball, allowing repetition. Containment of $a\mathbb{B}^D$ is closed in these coordinates. Every such P satisfies $U_b P \subseteq \mathbb{B}^D \subseteq a^{-1}P$. The containment conditions are closed on this compact set and $1 \leq \lambda \leq a^{-1}$; hence a minimum exists.

Let its vertices be v_s and λ its dilation. Initialize with a convex representation of ax . For each b, s , represent $\lambda^{-1}U_b v_s$ by the same vertices and use its convex coefficients as the row $T_b(s, \cdot)$. Induction gives

$$\mathbb{E}[v_{S_t} \mid w] = a\lambda^{-t}U_w x.$$

The decoder $d_j(s) = \rho\lambda^N \langle e_j, v_s \rangle / a$ lies in $[-1, 1]$ and gives (1.1). These are actual rows throughout the run, not independently chosen checkpoint factorizations. Carathéodory gives $D + 1$ successors; there are $|\mathcal{A}|k$ command rows, and their construction and storage remain uncharged.

The defining finite problem is semialgebraic: use convex coefficients for transformed vertices and a universal ball containment condition, then eliminate quantifiers. To prove (3.11), set $e = \max_b \|U_b - V_b\|_{\text{op}}$. For every feasible P , $V_b P \subseteq U_b P + e\mathbb{B}^D \subseteq (\lambda + e/a)P$. Interchanging U, V completes the proof. $\square$

The two profiles have different quantifiers. The converse tests every hidden machine; the enclosure theorem builds a particular positive realization. Neither theorem asserts that $\lambda_{\mathcal{A}}$ is the optimum over hidden machines. This is why finite-scale lower and upper bounds can be used even when they do not match.

Proposition 3.6 (Bounded word recoding). *Suppose every command of $\mathcal{A}$ is a product of exactly L commands of an alphabet $\mathcal{B}$ on the same space, with the same seeds and queries. Then*

$$\Gamma_{\mathcal{A}}(k, B) \leq \Gamma_{\mathcal{B}}(k, LB), \quad \lambda_{\mathcal{A}}(k, a) \leq \lambda_{\mathcal{B}}(k, a)^L, \quad W_{N,\varepsilon}(\mathcal{A}) \leq W_{LN,\varepsilon}(\mathcal{B}).$$

An identity letter permits words of length at most L to be padded. If both alphabets contain an identity and admit such bounded recodings into the other, the existence and value of a logarithmic width exponent agree.

Proof. Push forward a packet law along the word substitution to obtain the first inequality. Iterate the enclosure containments for the second. For the last inequality, compose the L stochastic rows of a $\mathcal{B}$ machine into one row for the corresponding $\mathcal{A}$ command, retaining only the block-end label. All rows may depend on the external epoch. This is an implementation of the atomic $\mathcal{A}$ interface, not a claim that its physical letters cost L letters. Applying the two comparisons and monotonicity under final identity padding proves the exponent statement. $\square$

Corollary 3.7 (Packing–dilation balance). *Fix a calibrated experiment, an inradius $a > \rho$, and positive numbers u, v . Suppose that for every sufficiently large k there are a length $B(k) \leq Ck^u$ and a word law for which*

$$\sup_{x,z \in \mathbb{S}^{D-1}} p\{w : \|U_w x - z\| < ck^{-v}\} \leq (2k)^{-1}.$$

Suppose also that $\lambda_{\mathcal{A}}(k, a) \leq \exp(C'k^{-u-2v})$. Then

$$W_{N,\varepsilon} = \Theta(N^{1/(u+2v)}).$$

The same conclusion holds with $u = 0$ when the lengths are uniformly bounded.

Proof. Proposition 3.4 gives $\Gamma(k, B(k)) \geq c^2 k^{-2v}/4$. Equation (3.8) implies $N \leq C_1 k^{u+2v}$ for a feasible peak k . Conversely choose k at a sufficiently large constant times $N^{1/(u+2v)}$ so that $C'Nk^{-u-2v} \leq \log(a/\rho)$, and apply Theorem 3.5. The hypotheses concern finite word orbits and compatible convex containments, and impose no commutativity on the matrices. $\square$

4. EQUAL-LENGTH WORDS AND BOUNDED REALIZATION

All groups in this section act faithfully on the active space R . Choose a reference command $A = U_{a_0}$. For $a \in \mathcal{A}$ and $t \in \mathbb{Z}$, write

$$d_a = A^{-1}U_a, \quad b_{t,a} = A^{-t}d_a A^t.$$

Lemma 4.1 (Tail generators and reference independence). *For every integer T , the closed subgroup generated by $\{b_{t,a} : t \geq T, a \in \mathcal{A}\}$ is $\mathcal{L}$. Equivalently, $\mathcal{L}$ is generated topologically by all equal-length differences $U_v^{-1}U_w$, $|v| = |w|$. It does not depend on the reference command.*

Proof. There are positive integers $n_i \rightarrow \infty$ with $A^{n_i} \rightarrow I$. For a finite-order matrix take multiples of the order; otherwise this follows by taking arbitrarily close pairs of powers in the compact closure of the cyclic subgroup. Consequently $b_{t+n_i,a} \rightarrow b_{t,a}$ for every integer t . Thus a closed tail-generated subgroup contains every $b_{t,a}$, $t \in \mathbb{Z}$. The closed subgroup L generated by these elements is invariant under conjugation by A and A^{-1} . It contains each d_a , and therefore is invariant under conjugation by every $U_a = Ad_a$ and its inverse. By continuity it is normal in H . Since $U_a^{-1}U_b = d_a^{-1}d_b$, it contains every generator in (1.2); conversely all the $b_{t,a}$ belong to $\mathcal{L}$. Hence $L = \mathcal{L}$.

Every $b_{t,a}$ with $t \geq 0$ is an equal-length difference: take the reference word of length $t+1$ and the word whose product is $U_a A^t$. On the other hand, all length- n word products have the same coset $A^n \mathcal{L}$ because the quotient identifies every command. All their differences lie in $\mathcal{L}$. These two inclusions prove the equivalent description and the reference independence. $\square$

The recurrence in the first paragraph of the proof can also be verified by simultaneous approximation of the finitely many eigenangles of A . No numerical recurrence rate is used.

Lemma 4.2 (Stabilization of normalized word orbits). *For a unit vector u in an H -invariant subspace, define*

$$F_t(u) = \{A^{-t}U_w u : w \in \mathcal{A}^t\}, \quad F_0(u) = \{u\}.$$

Then $F_t(u) \subseteq F_{t+1}(u)$. Their cardinalities are bounded in t if and only if the orbit $\mathcal{L}u$ is finite. More precisely, if $|F_t(u)| \leq k$ for all t , then $|\mathcal{L}u| \leq k$ and $\mathcal{L}^\circ u = u$.

Proof. Appending the reference command to a word gives the nesting. More generally

$$(4.1) \quad F_{t+1}(u) = \bigcup_{a \in \mathcal{A}} b_{t,a} F_t(u).$$

If the cardinalities are bounded, the increasing finite sets stabilize: $F_t(u) = F$ for all $t \geq T$, with $|F| \leq k$. Each $b_{t,a}$ maps F into F ; its invertibility makes that map a permutation. The group generated by the tail therefore preserves F . Its closure also preserves F , since F is finite and closed. Lemma 4.1 implies $\mathcal{L}u \subseteq F$. A connected group acts trivially on a finite orbit, proving the last assertion. Conversely $A^{-t}U_w u \in \mathcal{L}$, so a finite $\mathcal{L}$ -orbit contains every $F_t(u)$. $\square$

Lemma 4.3 (A positive packet on the moving subspace). *Let E be an H -invariant real subspace with $E^{\mathcal{L}^\circ} = \{0\}$. For every $k \geq 1$ there is a length $B_k \geq 1$ such that every unit $u \in E$ has*

$$|\{U_w u : w \in \mathcal{A}^{B_k}\}| > k.$$

For the uniform probability law p_k on all length- B_k words, the distortion of (3.1), computed in E , satisfies

$$(4.2) \quad \gamma_k := D_k^E(p_k) > 0.$$

Proof. The sets

$$C_t = \{u \in \mathbb{S}(E) : |F_t(u)| \leq k\}$$

are closed: for a finite collection of matrix products, the cardinality condition is a finite union of sets specified by vector equalities. They are nested by Lemma 4.2. If all were nonempty, compactness of $\mathbb{S}(E)$ would give a unit u in their intersection. The same lemma would give $\mathcal{L}^\circ u = u$, contrary to the hypothesis. Thus one C_{B_k} is empty. Multiplication by A^{B_k} does not change cardinality.

For a full-support word law, a scalar-product average in (3.1) can equal one only if every unit vector $U_w u$ equals one of the k centers. This is excluded for every u by the first assertion. Compactness of the direction and center spaces makes the maximum strictly less than one. This proves (4.2). This argument is qualitative: it supplies no dimension-free lower bound for γ_k . $\square$

Proof of Theorem 1.1. Suppose first that $\mathcal{L}$ is finite. Use labels $(x, \ell) \in X \times \mathcal{L}$, initialize at (x, I) , and at command epoch $t + 1$ update deterministically by

$$(4.3) \quad (x, \ell) \mapsto (x, b_{t,a}\ell).$$

An induction gives $\ell = A^{-t}U_w$ after the first t letters. The decoder at horizon N is

$$d_j(x, \ell) = \rho\langle e_j, A^N \ell x \rangle \in [-1, 1].$$

It gives (1.1) exactly. Only the label persists, and the time-dependent permutation in (4.3) uses the already declared external epoch. The held answer adds the maximum with two, not a free output buffer.

Now suppose $\mathcal{L}$ is infinite. Being a closed subgroup of a compact matrix group, it is a compact Lie group. A compact zero-dimensional Lie group is finite; hence its identity component is nontrivial. Let $E = R \ominus R^{\mathcal{L}^\circ}$. Normality of $\mathcal{L}^\circ$ in H makes both summands H -invariant and the

orthogonal projection P_E commute with all commands. If $E = 0$, $\mathcal{L}^\circ$ would act trivially on the faithful active space, a contradiction. If every seed had zero projection to E , its entire H -orbit would also, contradicting the definition of R . Therefore $\eta_X > 0$.

Choose a seed x attaining η_X and follow the unit orbit

$$Y_t = U_{a_t} \cdots U_{a_1} P_E x / \eta_X \in E.$$

The actual machine state is not changed or enlarged. For each complete word its decoder mean differs coordinatewise from $\rho U_w x$ by at most 2ε . Projection and pairing with Y_N give

$$\mathbb{E}\langle Y_N, P_E d(S_N) \rangle \geq \rho \eta_X - 2\varepsilon \sqrt{D}.$$

Since $\|P_E d\| \leq \sqrt{D}$, its conditional-centroid amplitude satisfies

$$(4.4) \quad q_N \geq \kappa_X := \rho \eta_X / \sqrt{D} - 2\varepsilon > 0.$$

This is one analytic pairing of the separate coordinate decoders, not a requirement to answer several queries jointly.

Fix a proposed peak k . Apply Lemma 4.3 on E , and use independent complete uniform packets of length B_k followed by a fixed deterministic remainder. Lemma 3.2 applies at their endpoints to the actual hidden register, regardless of its intermediate representations. On the remainder conditional Jensen cannot increase amplitude. Since $q_0 = 1$, every such machine must satisfy

$$(4.5) \quad \kappa_X \leq (1 - \gamma_k)^{\lfloor N/B_k \rfloor}.$$

If $\gamma_k = 1$, already one complete packet is excluded; otherwise $\lfloor N/B_k \rfloor [-\log(1 - \gamma_k)] \leq \log(1/\kappa_X)$. Thus this fixed k fails for every sufficiently large N . The conclusion is a limit to infinity, not merely an unbounded subsequence. Taking $\varepsilon = 0$ proves the exact converse. $\square$

Corollary 4.4 (A uniform basis-seed threshold). *For $X = \{\pm e_1, \dots, \pm e_D\}$ and fixed $\varepsilon < \rho/(2D)$, bounded width is equivalent to finiteness of $\mathcal{L}$. Otherwise $W_{N,\varepsilon} \rightarrow \infty$.*

Proof. If $E \neq 0$, then $\sum_j \|P_E e_j\|^2 = \dim E \geq 1$, so $\eta_X \geq D^{-1/2}$. Apply Theorem 1.1. $\square$

Corollary 4.5 (Every finite planar rotation alphabet). *For four coordinate seeds in the plane, commands with angles $2\pi\alpha_0, \dots, 2\pi\alpha_r$, and fixed $\varepsilon < \rho/(2\sqrt{2})$, width is bounded if and only if every $\alpha_i - \alpha_0$ is rational. If all differences have common denominator q , an exact machine uses at most $\max(2, 4q)$ labels. Otherwise width tends to infinity. The identity letter is not required.*

Proof. The commands commute, so $\mathcal{L}$ is the closed subgroup generated by their difference rotations. It is finite exactly in the rational-difference case. Otherwise it is the whole circle group, $E = \mathbb{R}^2$ and $\eta_X = 1$. The common rotation $N\alpha_0$ appears only in the horizon-dependent decoder of (4.3). $\square$

Remark 4.6 (Noncommuting examples and unused coordinates). Let $A_0 = R_\theta \oplus I_3$ on $\mathbb{R}^2 \oplus \mathbb{R}^3$, with $\theta/(2\pi)$ irrational, and use $A_0, R_\theta \oplus P, R_\theta \oplus Q$, where P, Q generate the six permutation matrices on three coordinates. The commands do not commute. Their active group is the infinite group $\text{SO}(2) \times S_3$, but $\mathcal{L} = \{I_2\} \times S_3$ is finite, giving at most sixty labels for the ten basis seeds. Conversely, if the active group is $\text{SO}(3)$ and two commands differ, its synchronization subgroup is $\text{SO}(3)$, since this group has no nontrivial proper closed normal subgroup. The width therefore diverges without importing a spectral-gap theorem. Finally, rotations on unused coordinates contribute nothing: for seeds on their common fixed axis, restriction to R makes the criterion finite. These examples distinguish synchronized action, ambient generated group, and seed activation.

5. A FINITE DECISION FOR ALGEBRAIC COMMAND DATA

The qualitative criterion does not require a spectral gap. For exact algebraic matrices it also has a finite decision procedure. Its termination uses a classical congruence bound for finite linear groups, not a truncation of an infinite harmonic norm.

Lemma 5.1 (A coarse finite-group bound). *If F is a number field of degree e and $G \subset \mathrm{GL}_d(F)$ is finite, then*

$$(5.1) \quad |G| \leq |\mathrm{GL}_{de}(\mathbb{F}_3)| < 3^{(de)^2}.$$

Proof. Restriction of scalars embeds G faithfully in $\mathrm{GL}_{de}(\mathbb{Q})$. For any finite subgroup of $\mathrm{GL}_n(\mathbb{Q})$ the lattice $\Lambda = \sum_{g \in G} g\mathbb{Z}^n$ is a G -invariant free abelian group of rank n . Choosing its basis conjugates G into $\mathrm{GL}_n(\mathbb{Z})$.

The kernel of reduction modulo three in $\mathrm{GL}_n(\mathbb{Z})$ has no nonidentity element of finite order. To prove this, a finite-order element would have a nonidentity power M of prime order p . Write $M = I + 3^s B$, $s \geq 1$, with some entry of B not divisible by three. If $p \neq 3$, expansion of $M^p = I$ modulo 3^{s+1} gives $pB \equiv 0 \pmod{3}$, a contradiction. If $p = 3$, divide the expansion by 3^{s+1} to obtain

$$B + 3^s B^2 + 3^{2s-1} B^3 = 0,$$

again impossible modulo three. Thus reduction is injective on the finite group. Finally $|\mathrm{GL}_n(\mathbb{F}_3)| = \prod_{j=0}^{n-1} (3^n - 3^j) < 3^{n^2}$. This is the standard invariant-lattice and congruence argument; no optimal Minkowski constant is needed. $\square$

Theorem 5.2 (Decidability of bounded exact width). *Suppose all command matrices and seeds have entries in a specified real number field F , with exact field arithmetic supplied. There is a terminating finite procedure deciding whether $\sup_N W_{N,0} < \infty$. It produces either a finite synchronization-group table and the exact bounded compiler, or a certificate that the synchronization group is infinite. No isotypic projection, connected-component oracle, or infinite spectral-gap certificate is an input.*

For symmetric coordinate seeds and any specified fixed $0 \leq \varepsilon < \rho/(2D)$, the same decision classifies bounded versus divergent $W_{N,\varepsilon}$.

Proof. Construct the active space by starting with the seed columns and repeatedly adjoining their images under all commands until the rank stops increasing. This takes at most D strict rank increases. Gaussian elimination over F supplies a basis of R of dimension d , and every restricted command is a matrix in $\mathrm{GL}_d(F)$. Rank over this real field agrees with rank over $\mathbb{R}$, since both are determined by the same nonzero minors. Stability under each invertible command also implies stability under its inverse.

Let A be one restricted reference command, and put $d_a = A^{-1}U_a|_R$. Start with a finite set S_0 containing I and all d_a . At each stage adjoin inverses, all pairwise products, and conjugates by A and A^{-1} . Equality is decided exactly in F . Stop if either

- (1) the finite set is closed under all these operations, or
- (2) its size exceeds $M = 3^{(de)^2}$.

Every generated element belongs to the algebraic group whose closure is $\mathcal{L}$. If the set closes, it is a finite subgroup normal under A , containing every d_a ; Lemma 4.1 shows that it is exactly $\mathcal{L}$. Its multiplication and conjugation tables define (4.3).

If its size exceeds M , Lemma 5.1 excludes a finite containing subgroup of $\mathrm{GL}_d(F)$. Hence $\mathcal{L}$ is infinite. The certificate is the list of more than M pairwise distinct matrices, together with their expression trees in the allowed operations. If neither stopping condition has occurred, the next stage strictly increases the finite cardinality. Thus after finitely many stages one stopping

condition must occur. Theorem 1.1 and Corollary 4.4 turn this group decision into the claimed width decision. $\square$

Remark 5.3 (Complexity and certificate size). The bound M is deliberately enormous; enumeration may be much worse than exponential in the input bit length, and intermediate algebraic entries may grow. This is a decision theorem, not an efficient implementation. Finite matrix-group algorithms provide more sophisticated approaches [15]; the elementary procedure above is included to prove termination with exactly the supplied data. In the infinite case it decides divergence, not its rate or the smallest excluding horizon for a proposed width.

Example 5.4. Take rational three-dimensional rotations a, b about two distinct coordinate axes with cosine $3/5$ and sine $4/5$, and alphabet $\{a, ab\}$. The difference $a^{-1}(ab) = b$ belongs to $\mathcal{L}$ and has infinite order: its rational trace is $11/5$, whereas the trace of a finite-order rational matrix is an integer, being both rational and a sum of roots of unity. Thus $\mathcal{L}$ is infinite. For basis seeds, the width diverges at every fixed error below $\rho/6$. This conclusion uses no numerical or qualitative gap theorem. A sharper linear growth law for this particular example still uses the independent expansion argument preserved in revision 42; it is not inferred from the decision alone.

6. HANKEL SECTIONS AND COMPATIBLE POSITIVE FACTORS

Fix a horizon N . For $0 \leq t \leq N$, define the probability prefix-suffix matrix

$$(6.1) \quad H_t[(x, u), (v, j, b)] = \frac{1 + b\rho\langle e_j, U_v U_u x \rangle}{2}, \quad |u| = t, \quad |v| = N - t.$$

The centered mean matrix has rank at most D , while $\text{rank } H_t \leq D + 1$. Each binary column pair is a conditional experiment, not a joint distribution over all suffixes or all queries. If a legal machine has K_t labels, conditioning on its state gives $H_t = E_t F_t$ with nonnegative factors, E_t row-stochastic and every (v, j) pair of columns in F_t summing to one. Let $\text{rank}_{\text{st}} H_t$ denote the minimum inner dimension under these stochastic normalizations. In particular

$$(6.2) \quad \text{rank } H_t \leq \text{rank}_+ H_t \leq \text{rank}_{\text{st}} H_t \leq K_t.$$

The last inequality does not reverse by choosing unrelated factorizations at different cuts. The same command row must transport all prefixes and all suffix responses simultaneously.

There is an exact formulation which preserves that compatibility. Let

$$C_t = [-1, 1]^{A^{N-t} \times \{1, \dots, D\}}$$

be the compact cube of possible terminal mean arrays. For $a \in \mathcal{A}$ let the continuation restriction be

$$(D_{t,a} f)(v, j) = f(av, j), \quad f \in C_t.$$

The concatenation av means execute a first and then v . The target root row is $r_x(w, j) = \rho\langle e_j, U_w x \rangle \in C_0$. A compatible polytope flag of profile $K_\bullet$ consists of nonempty $P_t \subset C_t$, each with at most K_t vertices, satisfying

$$(6.3) \quad D_{t,a} P_t \subset P_{t+1} \quad (0 \leq t < N, a \in \mathcal{A}).$$

This ambient continuation cube is generally much larger than the signed target representation. No hidden state is required to lie in the affine hull of the observed target rows.

Theorem 6.1 (Exact continuation-flag formula). *For a prescribed available profile $K_\bullet$, the minimum actual terminal row-TV error is*

$$(6.4) \quad \mathcal{E}_N(K_\bullet) = \frac{1}{2} \min_{(P_t) \text{ satisfying (6.3)}} \max_{x \in X} \text{dist}_\infty(r_x, P_0).$$

The minimum is attained. In particular exact realization is equivalent to a compatible flag with every $r_x \in P_0$. Available padding labels and zero-probability rows do not change the formula.

Proof. For any machine and each state s at cut t , let $f_{t,s}(v, j)$ be its actual future mean after executing suffix v and query j from that state. Every such vector belongs to C_t , including an unreachable state, since its rows are normalized. By the common command row,

$$D_{t,a}f_{t,s} = \sum_{s'} T_{t+1,a}(s, s') f_{t+1,s'}.$$

Thus $P_t = \text{conv}\{f_{t,s} : s \in S_t\}$ satisfies (6.3). Each realized initial row is a convex combination of the $f_{0,s}$, so the distance on the right of (6.4) cannot exceed twice the machine's error.

Conversely choose a flag and list its vertices, padding to K_t labels by repeats. For each vertex $f_{t,s}$ and command a , choose convex coordinates of $D_{t,a}f_{t,s}$ in P_{t+1} ; these coordinates are one stochastic row, shared by every suffix and every prefix reaching s . At cut N use the coordinates $f_{N,s}(j)$ as bounded decoder means. Backward induction shows that $f_{t,s}$ is the actual continuation vector of this machine. For each seed choose a nearest point of P_0 to r_x and initialize using its convex coordinates. The maximum mean error is exactly the corresponding maximum distance, and binary row-TV is half of mean error. This proves both inequalities.

For attainment, parametrize at most K_t vertices by K_t points in the compact cube, with stochastic convex weights for every inclusion and initialization. All inclusion equalities are closed polynomial constraints; the all-zero continuation flag is feasible. The error objective is continuous, so it attains a minimum. Equivalently, polytopes with at most K_t vertices form a compact family under Hausdorff distance in these finite cubes. Repeated vertices give the required padding convention. $\square$

This theorem is the finite controlled version of the classical shift-invariant-cone realization mechanism [12, 18]. Its role is to identify the exact resource and approximation objective. It does not claim a new discovery of Hankel representation or make nonnegative matrix factorization a convex global realization problem. For algebraic finite data, the displayed vertex and weight system is a finite real-algebraic optimization, with an exponential-size explicit suffix representation in general.

Proposition 6.2 (A dual row certificate, retained from revision 44). *For a candidate mean continuation c and next-cut mean generators f_i ,*

$$\min_{\theta \in \Delta} \left\| c - \sum_i \theta_i f_i \right\|_\infty = \max_{\|z\|_1 \leq 1} \left(\langle z, c \rangle - \max_i \langle z, f_i \rangle \right).$$

The value vanishes exactly when the proposed row has a stochastic extension. Initial mean error δ_0 and per-cut mean defects δ_t yield a machine with terminal binary-TV error at most $(\delta_0 + \sum_t \delta_t)/2$.

Proof. Express the infinity norm by duality with the compact unit ℓ^1 ball and apply finite bilinear minimax on that ball and the probability simplex. Minimization over the simplex selects a maximizing vertex for the subtracted term. The propagation bound follows because suffix restriction and stochastic convex averaging are contractions for the infinity norm. Binary total variation is half the mean error. Local errors may cancel, so this accumulated certificate need not be the actual optimum in Theorem 6.1. $\square$

Proposition 6.3 (Every planar cut has nonnegative rank three). *For any nonempty finite planar rotation alphabet, $X = \{\pm e_1, \pm e_2\}$ and $0 < \rho \leq 1/2$, each matrix H_t has*

$$\text{rank } H_t = \text{rank}_+ H_t = \text{rank}_{\text{st}} H_t = 3.$$

This includes the irrational-difference alphabets of Corollary 4.5, whose full-run widths diverge.

Proof. Choose an equilateral triangle of circumradius 2ρ centered at the origin. Its inradius is ρ , so every vector $\rho U_u x$ is a convex combination of its three vertices v_i . Use those coefficients for E_t .

Set

$$F_t[i, (v, j, b)] = \frac{1 + b\langle e_j, U_v v_i \rangle}{2}.$$

These entries are nonnegative because $\|v_i\| = 2\rho \leq 1$, and each binary pair sums to one. This proves the upper nonnegative-rank bound. For the lower bound, fix one prefix and one suffix. The four seed rows have affinely spanning planar means, and the two rotated coordinate queries span the dual plane. The binary complements supply the constant column. The resulting submatrix has rank three. Thus (6.2) is sharp at every cut, but not for simultaneous realization. $\square$

Proposition 6.4 (Small-signal finite-group compiler). *If $\mathcal{L}$ is finite on the active space of dimension d and $\rho \leq 1/d$, then all horizons admit exact width at most $\max\{2, (d+1)|\mathcal{L}|\}$, independently of the number of seeds.*

Proof. Take a regular unit-circumradius simplex in R , whose inradius is $1/d$. Its $d+1$ vertices and all their $\mathcal{L}$ translates form a finite set in the unit ball. Initialize ρx in the simplex by convex coordinates. A label (ℓ, i) represents ℓv_i in the rotating frame and is updated by the permutation $\ell \mapsto b_{t,a}\ell$. Its expected represented vector is $A^{-t}U_w \rho x$. The terminal column $\langle e_j, A^N \ell v_i \rangle$ is bounded by one and has exactly the desired mean. Duplicated orbit points may be merged but are not needed for the count. $\square$

7. A COMPARISON OF PACKET LOSS AND ENCLOSURE DILATION

Recall the distinct optimized quantities in Section 3. Their definitions do not imply equality with each other or with the hidden-state optimum. There is nevertheless a universal one-sided comparison, proved in revision 44 [22] and reproduced here. Define the packet loss rate

$$\mathfrak{r}_{\mathcal{A}}(k) = \sup_{B \geq 1} \frac{\chi_{\mathcal{A}}(k, B)}{B}.$$

Theorem 7.1 (Packet–enclosure comparison). *For every finite orthogonal alphabet, positive inradius a , and k for which the enclosure family is nonempty,*

$$(7.1) \quad \mathfrak{r}_{\mathcal{A}}(k) \leq \log \lambda_{\mathcal{A}}(k, a).$$

In particular, an invariant enclosure with at most k vertices and positive inradius implies $\Gamma_{\mathcal{A}}(k, B) = 0$ at every length B .

Proof. First take any feasible enclosure $P = \text{conv}(v_1, \dots, v_m)$, $m \leq k$, contained in the unit ball and containing $a\mathbb{B}$, with $U_a P \subset \lambda P$ for every command. Choose common stochastic rows representing $\lambda^{-1}U_a v_i$ by the vertices of P . For any fixed unit initial vector u , choose an initial distribution with mean au . For every word of length n the vertex mean is

$$\mathbb{E}[v_{S_n} \mid w] = a\lambda^{-n}U_w u.$$

Under any independent length- B packet law repeated j times, let $Y_{jB} = U_w u$. Then

$$a\lambda^{-jB} = \mathbb{E}\langle Y_{jB}, v_{S_{jB}} \rangle \leq \mathbb{E}\|\mathbb{E}[Y_{jB} \mid S_{jB}]\| = q_{jB}.$$

The equality is obtained by conditioning on the entire word; the inequality uses $\|v_s\| \leq 1$. Repeated application of Lemma 3.2 gives $q_{jB} \leq (1 - D_k(p))^j$. Taking j th roots and then $j \rightarrow \infty$ yields $\lambda^{-B} \leq 1 - D_k(p)$. Optimize over p , take logarithms and divide by B . Then optimize over B and take the infimum over feasible enclosures. No bounded final decoder at all these lengths was assumed; only the bounded auxiliary vertex column was used in the pairing. This avoids imposing an anytime response task. $\square$

The proof yields a lower bound for every common positive enclosure from a converse applicable to all hidden machines. It does not reconstruct such an enclosure from an arbitrary compatible flag in Theorem 6.1. The high-dimensional continuation cube in that theorem is exactly where additional hidden degrees of freedom may occur.

Proposition 7.2 (A geometric meaning of zero packet distortion). *For any k, B , the following are equivalent:*

- (1) $\Gamma_{\mathcal{A}}(k, B) = 0$;
- (2) *for one, hence every, full-support word law p on $\mathcal{A}^B$, $D_k(p) = 0$;*
- (3) *there is a unit u with $|\{U_w u : |w| = B\}| \leq k$.*

If $\Gamma_{\mathcal{A}}(k, B) = 0$ for every B , then a unit vector has a synchronization orbit of cardinality at most k . Conversely such a vector makes all those packet values zero.

Proof. Each scalar product in the full-support average is at most one. Equality of the maximum with one requires every orbit point to equal one of the centers, and attainment makes this an equivalence. The same set of centers then witnesses zero distortion for every word law. For the all-length assertion, use the nested compact sets C_B and Lemma 4.2. Conversely normalized word orbits lie in the finite synchronization orbit, and multiplication by the reference command preserves their cardinality. $\square$

An invariant direction may therefore make a whole-space packet profile zero even when active directions elsewhere require growing memory. The projection to $R \ominus R^{\mathcal{C}^o}$ in Theorem 1.1 is not cosmetic: it removes precisely this obstruction to using the packet profile for the boundedness problem.

8. RELATION TO POSITIVE REALIZATION AND STOCHASTIC AUTOMATA

For fixed seed and query, the centered response is a rational word series of signed linear dimension at most D . Positivity, normalization, and common continuation are additional requirements, not consequences of that dimension. The following comparisons locate those requirements before drawing any conclusion from the arithmetic examples.

8.1. Signed Hankel rank and positive order. Balle–Panangaden–Precup [14, Theorem 2] recall the classical equivalence between finite Hankel rank and minimum weighted-automaton dimension. It applies to signed rational series. In our notation it explains the first rank bound preceding (6.2), not the last inequality in the opposite direction. Their SVA truncation result [14, Theorem 11] concerns ℓ^2 error for absolutely summable rational series under a stability hypothesis. Our undamped orthogonal word series need not satisfy that hypothesis, and the approximating rows here must remain stochastic and all terminal columns bounded. A finite-length ℓ^2 error would bound a finite-length supremum error, but these positivity and stability conditions would still have to be established. We do not infer a stochastic compiler from signed minimization.

8.2. Positive cones and the ambient space. Benvenuti–Farina [12, Theorems 2–3] formulate positive realization through invariant polyhedral cones, including the shift representation. Their Section VII also explains why restricting a cone to a prescribed minimal observable realization can fail to recover the unrestricted minimum positive order. Theorem 6.1 is a finite, multi-command, normalized version of that mechanism. Its entire future-response cube is essential; replacing it by the D -dimensional target space would silently restrict the competing hidden states. The enclosure of Theorem 3.5 is a useful special cone section, not all such flags. Benvenuti’s later survey [13] provides further background on minimum positive order.

Stationary realization asks for one unchanged set of transition and output data at every length. The present quantifiers are instead “for every N there exists a layered machine”, with the final

decoder depending on N . The one-letter irrational example is a concrete distinction, rather than an assertion that all classical stationary lower bounds transfer. Likewise Proposition 6.3 shows that even every separate positive factorization can have size three while the compatible full-run width diverges.

8.3. Stochastic realization and factorization consistency. Heller’s finite stochastic-realization framework [18] is part of the classical lineage of shift-invariant positive cones. Ohta [20, Theorems 6–9 and Remark 10] analyzes reachable and null spaces, effective dimension, tensor decomposition and generalized Hankel matrices for stationary hidden Markov models. These distinguish observable dimension from a particular hidden presentation. In our controlled model, each command is externally specified and each row is stochastic separately; an emission-indexed HMM kernel instead has a different normalization across its emissions. Neither a stationary effective-space reduction nor a tensor-rank bound alone supplies (6.3) for every command and epoch.

Finesso–Grassi–Spreij [16, Section IV] explicitly separate pseudo-Hankel nonnegative factorization from the subsequent parametrization step needed to obtain HMM transition data. Their objective uses informational divergence of finite distributions. This is directly relevant to the warning after (6.2): independent factors are not yet a common stochastic dynamics. Our exact flag formula uses worst-word binary total variation, equivalent to one half the supremum error of means. It is a finite characterization, not a convex solution of the generally nonconvex factorization problem.

8.4. What the boundedness criterion adds. The coset obstruction in group random walks is classical; for the finite-group aperiodicity statement, see the introduction of McCarthy [19]. The subgroup in (1.2) is the smallest closed normal subgroup in whose coset all commands lie. We do not claim that this group-theoretic object, invariant cones, or finite-group recognition was first introduced here. Our conclusion is the equivalence of its finiteness on the active space with uniformly bounded horizon-specific *numerical* positive width. Its converse combines equal-length orbit stabilization, compact uniformization on the moving subspace, and the inherited arbitrary-hidden-state packet inequality. It applies without a gap, with noncommuting commands, and across disconnected actions.

Finite matrix-group recognition is an established algorithmic subject. Detinko–Flannery–O’Brien [15, Sections 3.1–3.2 and 4] give much stronger computational methods than our coarse enumeration. Theorem 5.2 uses a proved order bound to close the normal-conjugation enumeration and turns the new memory criterion into a finite algebraic decision. It is neither a new general finiteness algorithm nor an infinite spectral-gap oracle.

The arithmetic results below retain their original distinctions: fixed alphabet size, fixed calibration, and logarithmic exponents rather than constant-factor laws where only ordinary Diophantine type is assumed. Their standard individual versus uniform exponent convention is made explicit using German [17, Definitions 1–2]: the former asks for infinitely many approximations, whereas the latter asks for one at every sufficiently large scale. The row-vector dual exponent in (9.1) is the individual exponent in that convention; changing from the supremum norm to ℓ^1 does not change its value. The direct DGLPS comparison is retained in the final arithmetic discussion.

9. RETAINED ARITHMETIC THEOREMS

The following results and their proofs are retained from revision 43 [21]. Constants may depend on the fixed angle vector, the alphabet size, signal, tolerance, and displayed logarithmic parameter; they are not uniform near rational relations or as the alphabet size grows. Our arithmetic conclusions go beyond uniform bad approximation. For $\alpha \in \mathbb{R}^r$ with $1, \alpha_1, \dots, \alpha_r$ rationally

independent, write

$$(9.1) \quad \omega^*(\alpha) = \limsup_{\|m\|_1 \rightarrow \infty} \frac{-\log \|m \cdot \alpha\|_{\mathbb{R}/\mathbb{Z}}}{\log \|m\|_1}.$$

The alphabet consists of the identity and the rotations through $2\pi\alpha_i$. Throughout, r , the alphabet, $0 < \rho \leq 1/10$ and $0 \leq \varepsilon < \rho/(2\sqrt{2})$ are fixed.

Theorem 9.1 (Minimal Diophantine type). *If $\omega^*(\alpha) = r$, then*

$$(9.2) \quad \lim_{N \rightarrow \infty} \frac{\log W_{N,\varepsilon}(\alpha)}{\log N} = \frac{r}{2r+1}.$$

For Lebesgue-almost every α , and every fixed $\delta > 0$, there are positive constants c, C such that, for $N \geq 2$,

$$(9.3) \quad c \frac{N^{r/(2r+1)}}{(\log(N+2))^{2r(1+\delta)/(2r+1)}} \leq W_{N,\varepsilon}(\alpha) \leq CN^{r/(2r+1)} (\log(N+2))^{2r^2(1+\delta)/(2r+1)}.$$

The upper inequalities are achieved by exact common-row machines. The constants in the almost-everywhere bounds may depend on the fixed $\alpha, \delta, r, \rho, \varepsilon$; they are not uniform near rational relations or for growing r .

The exponent in (9.2) need not imply a two-sided constant-factor law. The earlier dual badly approximable theorem supplies the stronger Θ conclusion on its stated class, and is recovered in Corollary 11.4. Theorem 9.1 instead allows arbitrarily small Diophantine constants along subsequences. The quantitative arithmetic ingredients are classical; the memory conclusions concern their compatible realization and all-hidden-state converse.

There is also a genuine failure of a single exponent, even for one fixed irrational angle.

Theorem 9.2 (Irrational horizons and Liouville fluctuations). *For every irrational α there are horizons $M_n \rightarrow \infty$ such that $W_{M_n,\varepsilon}(\alpha) = \Theta(M_n^{1/3})$. If α is Liouville, then*

$$(9.4) \quad \liminf_{N \rightarrow \infty} \frac{\log W_{N,\varepsilon}(\alpha)}{\log N} = 0, \quad \frac{1}{3} \leq \limsup_{N \rightarrow \infty} \frac{\log W_{N,\varepsilon}(\alpha)}{\log N} \leq \frac{1}{2}.$$

Nevertheless $W_{N,\varepsilon}(\alpha) \rightarrow \infty$. All displayed upper bounds have exact realizations.

The upper limit in (9.4) is not determined here. The result establishes oscillation between subpower subsequences and a positive-power subsequence; it does not replace that distinction by a guessed exponent for all irrationals.

10. EXACT OPTIMIZATION OF PLANAR PACKETS

For a rotation alphabet, let $z_1, \dots, z_M \in \mathbb{S}^1 \subset \mathbb{C}$ be the distinct product phases of words of length B , ordered cyclically. A common rotation of all phases can be absorbed into the decoder centers. The initial direction in (3.1) therefore disappears. Any probability distribution on these phases is implemented by a distribution on representative words of the same length.

Let $\mathfrak{P}_k$ be the finite set of partitions of the cyclically ordered list into at most k nonempty consecutive arcs. For $p \in \Delta_M$, set

$$(10.1) \quad F_k(p) = \max_{\mathcal{P} \in \mathfrak{P}_k} \sum_{A \in \mathcal{P}} \left| \sum_{i \in A} p_i z_i \right|.$$

The decoder centers in this section are unit vectors on the circle; hence maximizing their scalar product is exactly nearest-center chordal quantization.

Theorem 10.1 (Optimal word laws as a cone program). *For the planar alphabet above,*

$$(10.2) \quad \Gamma_{\mathcal{A}}(k, B) = 1 - \min_{p \in \Delta_M} F_k(p).$$

Equivalently, minimize t subject to $p \in \Delta_M$ and, for every $\mathcal{P} \in \mathfrak{P}_k$,

$$(10.3) \quad \left\| \sum_{i \in A} p_i (\Re z_i, \Im z_i) \right\|_2 \leq r_{\mathcal{P}, A} \quad (A \in \mathcal{P}), \quad \sum_{A \in \mathcal{P}} r_{\mathcal{P}, A} \leq t.$$

This is a finite second-order cone program with attained optimum. A fixed law can be evaluated by cyclic segmentation in $O(kM^3)$ arithmetic operations, including computation of all interval sums. The optimization over laws may have exponentially many partition constraints.

Proof. For fixed centers $v_1, \dots, v_k$ assign each phase to a nearest center in chordal distance; equivalently maximize $\Re(\bar{v}_j z_i)$. Distinct centers have connected cyclic Voronoi arcs, bounded by the angular midpoints of adjacent centers. Ties can be assigned consecutively; coincident centers can be merged. For the induced partition, the contribution of a block A is at most $|\sum_{i \in A} p_i z_i|$.

Conversely, for any cyclic partition choose the center of each block in the direction of its weighted complex sum, arbitrarily if that sum vanishes. The best assignment to those centers has value at least the sum in (10.1). Thus the maximum over centers equals $F_k(p)$. This proves (10.2); introducing the norm epigraphs proves (10.3). Zero weights follow either directly or by continuity from positive weights.

For a fixed cyclic cut, prefix sums give every block vector in constant time. The usual recurrence for the best j blocks among the first m points checks at most M previous split positions. It costs $O(kM^2)$ for that cut. Repeating for the M cuts costs $O(kM^3)$. This recurrence evaluates a given law; it is not a polynomial-size representation of all constraints in (10.3). Compactness of the simplex and continuity give attainment. $\square$

Nearest-center partitions and vector quantization are classical [5]. The additional optimization here chooses the distribution of actual command words, and Theorem 3.3 turns its value into a constraint on arbitrary hidden registers. There is no assertion that static quantization alone characterizes the least whole-run width.

Proposition 10.2 (A finite exact packet certificate). *For the alphabet consisting of identity and a quarter turn, at word length three the four distinct phases are $1, i, -1, -i$. Their uniform law is optimal and*

$$\Gamma(1, 3) = 1, \quad \Gamma(2, 3) = 1 - \frac{1}{\sqrt{2}}, \quad \Gamma(3, 3) = \frac{2 - \sqrt{2}}{4}, \quad \Gamma(k, 3) = 0 \quad (k \geq 4).$$

Proof. The distortion is concave in the phase law and invariant under its four cyclic rotations. Averaging any law over those rotations proves optimality of the uniform law. For two centers, adjacent pairs give total $\sqrt{2}/2$; a triple and a singleton give only $1/2$. For three centers, one pair and two singletons give $\sqrt{2}/4 + 1/2$. For one center the sum is zero; four centers fit all phases. Formula (10.2) gives the values. A uniform phase law is realized by the four ordinary words with zero, one, two or three initial quarter turns and then idle commands. $\square$

For algebraic rotation matrices, both the general quantifier-elimination formulation and the cone program have algebraic coefficients. No finite list of spherical harmonics is used. For merely computable-real entries, finite numerical approximations are diagnostics unless a rigorous error enclosure is supplied; (3.3) then quantifies how an entry enclosure propagates to a packet certificate.

Corollary 10.3 (Separated phases). *If $M \geq 2k$ distinct length- B word phases have pairwise circular distance at least $s \in (0, 1/2]$, then $\Gamma_{\mathcal{A}}(k, B) \geq s^2$.*

Proof. Use their uniform law. Each open arc of circular radius $s/2$ contains at most one phase. At least half the mass is at distance at least $s/2$ from every one of k centers. On $[0, 1/2]$, $1 - \cos(2\pi t) = 2\sin^2(\pi t) \geq 8t^2$. The average loss is at least s^2 . This is a lower certificate for (10.2), not necessarily its exact value. $\square$

11. ARITHMETIC PROFILES AT EVERY FINITE SCALE

Now $\mathcal{A}_\alpha = \{I, R_{2\pi\alpha_1}, \dots, R_{2\pi\alpha_r}\}$, with r fixed. Define

$$(11.1) \quad \psi_\alpha(H) = \min_{0 < \|m\|_1 \leq H} \|m \cdot \alpha\|_{\mathbb{R}/\mathbb{Z}}, \quad \eta_\alpha(q) = \max_{1 \leq i \leq r} \|q\alpha_i\|_{\mathbb{R}/\mathbb{Z}}.$$

These are finite functions of the alphabet parameters. The first may vanish in the presence of a rational relation. The upper profile remains usable in that case.

Theorem 11.1 (Finite arithmetic bounds). *Let $\kappa = \rho/\sqrt{2} - 2\varepsilon > 0$. For a machine of peak K , put $L = \lceil (2K)^{1/r} \rceil$ and $B = r(L-1)$. Then*

$$(11.2) \quad \lfloor N/B \rfloor \psi_\alpha(B)^2 \leq \log(1/\kappa).$$

For every integer $q \geq 4$,

$$(11.3) \quad N\eta_\alpha(q) \leq q^2/100 \implies W_{N,0}(\alpha) \leq q.$$

Consequently every finite planar rotation alphabet satisfies

$$(11.4) \quad W_{N,0}(\alpha) \leq \max(4, \lceil \sqrt{50N} \rceil).$$

All upper bounds use common command rows and hold also after adjoining reflection in the first coordinate axis.

Proof. Construct a word of length B in r consecutive segments of $L-1$ positions. In segment i give J_i copies of the i th rotation followed by idle commands, independently and uniformly for $J_i \in \{0, \dots, L-1\}$. Differences of count vectors have ℓ_1 norm at most B . If $\psi_\alpha(B) > 0$, the $L^r \geq 2K$ phases are distinct and separated by $\psi_\alpha(B)$. Corollary 10.3 and (3.8), with $-\log(1-x) \geq x$, give (11.2). If the minimum is zero the displayed inequality is automatic. All letters in a segment are actual commands, not a free count input.

For the upper bound let P_q be the regular q -gon inscribed in the unit circle. Choose nearest integers p_i to $q\alpha_i$, and put $\delta_i = 2\pi(\alpha_i - p_i/q)$, so $|\delta_i| \leq \pi/q$. Elementary chord geometry gives

$$(11.5) \quad R_{2\pi\alpha_i}P_q \subseteq \Lambda_i P_q, \quad \Lambda_i = \frac{\cos(\pi/q - |\delta_i|)}{\cos(\pi/q)}, \quad \log \Lambda_i \leq \frac{4\pi^2 \eta_\alpha(q)}{q^2}.$$

Indeed the ray of a rotated vertex meets a neighboring edge at radial coordinate $\cos(\pi/q)/\cos(\pi/q - |\delta_i|)$. Integrating the derivative of $\log \cos(\pi/q - t)$ bounds the logarithm by $|\delta_i| \tan(\pi/q)$, and $\tan(\pi/q) \leq 2\pi/q$ for $q \geq 4$.

Take $\Lambda = \max_i \Lambda_i$, including one for the identity. Under (11.3), $\Lambda^N \leq e^{\pi^2/25} < 2$. Since P_q contains the half-unit disk, Theorem 3.5 with $a = 1/2$ applies: $\rho\Lambda^N \leq 1/5 < 1/2$. Explicitly, initialize at mean $x/2$, represent $\Lambda^{-1}R_{2\pi\alpha_i}v_s$ by the vertices for each row, and decode by $2\rho\Lambda^N \langle e_j, v_s \rangle$. This verifies every numerical word, including any deterministic remainder of a converse packet decomposition.

There are $(r+1)q$ common command rows and two decoder columns, each row having at most three successors. These counts exclude row-description and arithmetic costs. Since $\eta_\alpha(q) \leq 1/2$, (11.4) follows. Reflection preserves P_q ; its contracted image uses the same vertex set. The reflected lower bound is obtained simply by restricting to rotation-only words. $\square$

The pair $(\psi_\alpha, \eta_\alpha)$ is a specialization of the word-distortion and enclosure profiles. It is not assumed to characterize exact finite width. In particular, (11.2) can be weak when long word packets have clusters; the optimized profile of Section 10 is available without that separation relaxation.

Lemma 11.2 (Denominators under a general power bound). *Suppose for $\tau \geq r$ and $b > 0$,*

$$(11.6) \quad \|m \cdot \alpha\|_{\mathbb{R}/\mathbb{Z}} \geq b \|m\|_1^{-\tau} \quad (m \neq 0).$$

For every integer $Q \geq 2$ there is q with

$$(11.7) \quad cQ^{r/(\tau+1-r)} \leq q \leq Q^r, \quad \eta_\alpha(q) \leq Q^{-1}, \quad c = (b/r^{\tau+1})^{r/(\tau+1-r)}.$$

Proof. The $Q^r + 1$ points $j\alpha$ in Q^r equal torus boxes give integers p_i, q with $1 \leq q \leq Q^r$ and $|q\alpha_i - p_i| \leq Q^{-1}$. For $M = \lfloor q^{1/r} \rfloor$, the $(M+1)^r > q$ integer vectors in $\{0, \dots, M\}^r$ have a residue collision for $m \cdot p \pmod{q}$. Subtracting gives $m \neq 0$ with $H = \|m\|_1 \leq r q^{1/r}$ and $q \mid m \cdot p$. Thus

$$bH^{-\tau} \leq \|m \cdot \alpha\|_{\mathbb{R}/\mathbb{Z}} \leq H/(qQ), \quad bqQ \leq r^{\tau+1} q^{(\tau+1)/r}.$$

Rearranging yields (11.7). This direct residue argument supplies exactly the quantitative transference needed here, without an unspecified transference constant. $\square$

Corollary 11.3 (Power bounds at finite Diophantine type). *Under (11.6), for constants depending on the fixed data,*

$$(11.8) \quad c_1 N^{r/(2\tau+1)} \leq W_{N,\varepsilon}(\alpha) \leq C_1 N^{\min\{1/2, r(\tau+1-r)/(\tau+1+r)\}}.$$

No matching claim is made when $\tau > r$.

Proof. From (11.2), $N < B + b^{-2} B^{2\tau+1} \log(1/\kappa)$, while $B \leq r(2K)^{1/r}$. This gives the lower power after adjusting finitely many initial horizons. Lemma 11.2 bounds $N\eta_\alpha(q)/q^2 \leq c^{-2} N Q^{-1-2r/(\tau+1-r)}$. Choose Q at constant times $N^{(\tau+1-r)/(\tau+1+r)}$, large enough to satisfy (11.3) and $q \geq 4$. Then $q \leq Q^r$ proves the second candidate upper power; (11.4) proves the first. $\square$

Corollary 11.4 (The retained badly approximable law). *If (11.6) holds with $\tau = r$, then $W_{N,\varepsilon}(\alpha) = \Theta(N^{r/(2r+1)})$, also with the reflection extension. This is the previously established matched law [8].*

Proof. Substitute $\tau = r$ in (11.8); the two powers coincide. $\square$

For $1, \alpha_1, \dots, \alpha_r$ rationally independent, every optimized finite-block action gap on the circle is zero. Indeed simultaneous approximation supplies unbounded frequencies n with every $n\alpha_i$ close to an integer. Each fixed rotation word then acts almost identically on its Fourier mode. For the reflected alphabet use the real cosine mode, which reflection fixes. This is the established almost-invariance argument, not a claim that the finite packet distortion vanishes.

12. MINIMAL ORDINARY TYPE AND ALMOST-EVERYWHERE WIDTHS

Uniform bad approximation is substantially stronger than the condition needed to determine the logarithmic width exponent. We first deduce the minimal-type assertion without any measure-theoretic hypothesis.

Proof of (9.2) in Theorem 9.1. If $\omega^*(\alpha) = r$, then for every $0 < h < 1/(2r-1)$ there is $b_h > 0$ such that

$$\|m \cdot \alpha\|_{\mathbb{R}/\mathbb{Z}} \geq b_h \|m\|_1^{-r-h} \quad (m \neq 0).$$

Indeed the definition of the limsup gives this inequality for all sufficiently large $\|m\|_1$, and rational independence makes the finite remaining minimum positive. Apply Corollary 11.3 with $\tau = r + h$. It gives

$$\frac{r}{2r + 2h + 1} \leq \liminf_N \frac{\log W_{N,\varepsilon}}{\log N} \leq \limsup_N \frac{\log W_{N,\varepsilon}}{\log N} \leq \min \left\{ \frac{1}{2}, \frac{r(1+h)}{2r+1+h} \right\}.$$

The second term is the minimum because $h(2r-1) < 1$. Let $h \downarrow 0$. This proves (9.2). No positive uniform constant at exponent r was used. $\square$

For completeness we prove a more quantitative metric statement, using only the summable part of metric Diophantine approximation. It does not claim a new metric theorem.

Lemma 12.1 (A summable logarithmic separation). *For Lebesgue-almost every $\alpha \in [0, 1]^r$, and every $s > 1$, there is $b_s > 0$ such that*

$$(12.1) \quad \|m \cdot \alpha\|_{\mathbb{R}/\mathbb{Z}} \geq \frac{b_s}{\|m\|_1^r (\log(2 + \|m\|_1))^s} \quad (m \neq 0).$$

Such a vector has $\omega^*(\alpha) = r$.

Proof. For a nonzero integer vector m , the map $\alpha \mapsto m \cdot \alpha \pmod{1}$ sends Haar measure on the torus to uniform circle measure. There are at most $C_r H^{r-1}$ integer vectors with $\|m\|_1 = H$. Hence the measures of the events

$$\|m \cdot \alpha\|_{\mathbb{R}/\mathbb{Z}} < [\|m\|_1^r (\log(2 + \|m\|_1))^s]^{-1}$$

have finite sum, bounded by a constant times $\sum_{H \geq 1} [H(\log(2 + H))^s]^{-1}$. The measure of their tail union tends to zero. Almost every vector violates the unscaled inequality only finitely often. Remove the measure-zero rational hyperplanes and reduce the constant to cover the finite exceptions. Intersecting the full-measure sets for rational $s > 1$ gives the assertion for every real $s > 1$: choose rational $1 < s_0 < s$ and use $(\log(2 + H))^{-s} \leq (\log(2 + H))^{-s_0}$.

This bound implies $\omega^*(\alpha) \leq r$. For the reverse inequality, place the $(M+1)^r$ values $m \cdot \alpha \pmod{1}$, $m \in \{0, \dots, M\}^r$, in fewer than $(M+1)^r$ intervals of equal length. A collision gives a nonzero difference of norm at most rM and distance $O(M^{-r})$. As $M \rightarrow \infty$ the norms of some such differences must tend to infinity, since rational independence excludes an exact relation. Thus $\omega^*(\alpha) \geq r$. $\square$

Proposition 12.2 (Quantitative bounds under logarithmic separation). *Suppose (12.1) holds for some $s > 0$ and $b_s > 0$. Then there are $c, C > 0$ such that*

$$(12.2) \quad cN^{r/(2r+1)} (\log(N+2))^{-2rs/(2r+1)} \leq W_{N,\varepsilon}(\alpha) \leq CN^{r/(2r+1)} (\log(N+2))^{2r^2s/(2r+1)}.$$

Proof. The packet bound (11.2) gives

$$N < B + b_s^{-2} B^{2r+1} (\log(2 + B))^{2s} \log(1/\kappa).$$

With $B \leq r(2K)^{1/r}$ this implies $N \leq C_0 K^{(2r+1)/r} (\log(2 + K))^{2s}$. If $K > N$, the proposed lower bound is already true after reducing its constant. Otherwise $\log(2 + K) \leq \log(2 + N)$, and rearrangement proves the first inequality of (12.2).

For the upper inequality repeat the residue argument of Lemma 11.2. It produces $q \leq Q^r$, $\eta_\alpha(q) \leq Q^{-1}$ and $H \leq rq^{1/r}$ with

$$\frac{b_s}{H^r (\log(2 + H))^s} \leq \frac{H}{qQ}.$$

Since $q \leq Q^r$, it follows for $Q \geq 2$ that

$$(12.3) \quad q \geq c_0 \frac{Q^r}{(\log(Q+2))^{rs}}.$$

Indeed $b_s q Q \leq H^{r+1}(\log(2+H))^s \leq C_r q^{1+1/r}(\log(Q+2))^s$, and raising the rearranged inequality to the r th power gives (12.3).

Choose an integer

$$Q = \left\lceil C_2 N^{1/(2r+1)} (\log(N+2))^{2rs/(2r+1)} \right\rceil.$$

For sufficiently large fixed C_2 , (12.3) implies $N\eta_\alpha(q)/q^2 \leq 1/100$ and $q \geq 4$. To see the constant choice explicitly, for large N one has $\log(Q+2) \leq C_3 \log(N+2)$ with C_3 fixed; thus the ratio is bounded by a constant times $C_2^{-(2r+1)}$. Enlarge the final constant for the finitely many initial horizons. Theorem 11.1 gives an exact machine with $q \leq Q^r$ labels, proving the upper bound. $\square$

Completion of Theorem 9.1. Apply Lemma 12.1 with $s = 1 + \delta$ and then Proposition 12.2. This yields (9.3) on one full-measure set for all fixed $\delta > 0$. These statements concern a fixed alphabet before the horizon grows. None of their constants is asserted uniform in r or near a rational relation. $\square$

12.1. Rational dependencies without redundant exponents. The number of listed rotations need not be the relevant arithmetic rank. The following result covers finite rational reparametrizations rather than assigning one exponent to every redundant letter.

Corollary 12.3 (Finite rational extensions). *Suppose an alphabet contains I and rotations through angles $\beta_1, \dots, \beta_d$, where $1, \beta_1, \dots, \beta_d$ are rationally independent and $\omega^*(\beta) = d$. Suppose every additional angle belongs to $\mathbb{Q} + \sum_{j=1}^d \mathbb{Q}\beta_j$. Then*

$$\lim_N \frac{\log W_{N,\varepsilon}}{\log N} = \frac{d}{2d+1}.$$

The same assertion holds on a full-measure set of β with the logarithmic upper and lower powers of (9.3), replacing r by d . Reflection may be added. If all angles are rational, exact width is bounded independently of N .

Proof. Restriction to the displayed d -angle subalphabet gives the lower bound. Choose one common denominator h and integers t_i, n_{ij} so that every angle is $t_i/h + \sum_j n_{ij}\gamma_j$ with $\gamma_j = \beta_j/h$. Multiplication by h shows $\|m \cdot \gamma\|_{\mathbb{R}/\mathbb{Z}} \geq h^{-1} \|m \cdot \beta\|_{\mathbb{R}/\mathbb{Z}}$. Thus every separating power or logarithmic inequality for β holds for γ with a changed constant.

Take a simultaneous approximant p_j/q to γ from the proof above. A command's irrational part has residual angle at most $2\pi(\sum_j |n_{ij}|)/(qQ)$. The same polygon construction therefore works, after changing only fixed constants. Store in addition the rational phase $c \in \mathbb{Z}/h\mathbb{Z}$, updated by $c \mapsto c + t_i$. The decoder rotates the stored polygon vector by $2\pi c/h$. This uses a peak of $\max(2, hq)$ labels, not an uncharged phase. Reflection sends c to $-c$ and conjugates the polygon phase; the regular polygon and the construction are reflection-invariant. The exact expectation identity is unchanged. The finite multiplier h does not change a logarithmic exponent or the displayed logarithmic powers.

For a purely rational alphabet, choose a common period h and retain the seed identifier and the accumulated phase. This uses at most $4h$ command labels for our four planar seeds, plus the held binary answer convention. No infinite-gap hypothesis is involved. $\square$

13. EVERY IRRATIONAL ANGLE HAS CUBIC SCALES

We keep the two-letter alphabet $\{I, R_{2\pi\alpha}\}$ and the planar resource and error conventions. No Diophantine lower bound is assumed in this section. The needed continued-fraction estimates are classical [3]; we include their short proof to specify the exact finite inequalities being used.

Lemma 13.1 (Convergent separation). *Let p_n/q_n be the convergents of an irrational α , ignoring the finitely many initial indices with repeated denominator. Then*

$$(13.1) \quad \frac{1}{q_{n+1} + q_n} < |q_n \alpha - p_n| < \frac{1}{q_{n+1}}, \quad \min_{1 \leq l < q_n} \|l\alpha\|_{\mathbb{R}/\mathbb{Z}} > \frac{1}{q_n + q_{n-1}} > \frac{1}{2q_n}.$$

Proof. The convergent recurrences imply $p_n q_{n-1} - p_{n-1} q_n = \pm 1$ and alternating signs of $e_n = q_n \alpha - p_n$. If ξ is the next complete quotient, then

$$\alpha = \frac{p_n \xi + p_{n-1}}{q_n \xi + q_{n-1}}, \quad a_{n+1} < \xi < a_{n+1} + 1.$$

It follows that $|e_n| = 1/(q_n \xi + q_{n-1})$, proving the first pair.

For $0 < l < q_n$ and an integer p , the determinant identity writes $(p, l) = a(p_n, q_n) + b(p_{n-1}, q_{n-1})$ with integers a, b . Here $b \neq 0$; a, b cannot both be positive, and cannot both be negative. If $a = 0$ the error has modulus at least $|e_{n-1}|$. Otherwise they have opposite signs, so ae_n and be_{n-1} have the same sign and again $|l\alpha - p| \geq |e_{n-1}|$. Minimize over p and apply the first pair with index $n - 1$. This proves the remaining inequalities. $\square$

Theorem 13.2 (Universal cubic subsequences). *Set $\kappa = \rho/\sqrt{2} - 2\varepsilon > 0$, $C_\kappa = 8(1 + \log(1/\kappa))$, and*

$$M_n = \lceil C_\kappa q_n^3 \rceil, \quad m_\kappa = \lceil 100(C_\kappa + 1) \rceil.$$

For all sufficiently large n ,

$$(13.2) \quad q_n/2 < W_{M_n, \varepsilon}(\alpha) \leq m_\kappa q_n.$$

The upper bound is exact, uses common command rows and requires no regularity of the partial quotients.

Proof. Use words of length $B = q_n - 1$ whose number of initial active letters ranges uniformly over $0, \dots, q_n - 1$. Lemma 13.1 gives q_n distinct phases with separation greater than $1/(2q_n)$. If the proposed peak K were at most $q_n/2$, Corollary 10.3 and the packet budget would give

$$\left\lfloor \frac{M_n}{q_n - 1} \right\rfloor \frac{1}{4q_n^2} \leq \log(1/\kappa).$$

The left side is at least $C_\kappa/4 - 1/(4q_n^2)$, strictly greater than the right side. This proves the lower bound against all hidden-state machines.

For the upper bound use the polygon denominator $Q_n = m_\kappa q_n$. Distance to the nearest integer is subadditive, so

$$\|Q_n \alpha\|_{\mathbb{R}/\mathbb{Z}} \leq m_\kappa |q_n \alpha - p_n| < m_\kappa / q_{n+1} \leq m_\kappa / q_n.$$

Consequently $Q_n^2 / [100 \|Q_n \alpha\|_{\mathbb{R}/\mathbb{Z}}] > m_\kappa q_n^3 / 100 \geq (C_\kappa + 1) q_n^3 \geq M_n$. Equation (11.3) gives the exact upper construction. The constants have been chosen for transparency, not optimality. $\square$

Proposition 13.3 (Resonant long horizons). *For any irrational α and integer $q \geq 4$ with $N_q = \lfloor q^2 / (100 \|q\alpha\|_{\mathbb{R}/\mathbb{Z}}) \rfloor \geq 1$, one has $W_{N_q, 0}(\alpha) \leq q$. If α is Liouville, there is a sequence $N \rightarrow \infty$ on which $\log W_{N, \varepsilon}(\alpha) / \log N \rightarrow 0$.*

Proof. The first assertion is the finite enclosure implication (11.3). For a Liouville number choose arbitrarily large A and unbounded denominators q with $0 < \|q\alpha\|_{\mathbb{R}/\mathbb{Z}} < q^{-A}$. The first assertion then supplies horizons $N_q \geq q^{A+2}/200$ for large q , with width at most q . The logarithmic ratio is at most $1/(A+2) + o(1)$. A diagonal sequence with $A \rightarrow \infty$ proves the claim, since widths are at least two. $\square$

Proof of Theorem 9.2. Equation (13.2) and $M_n = \Theta(q_n^3)$ give the first assertion and a subsequence of logarithmic exponent $1/3$. Proposition 13.3 gives the zero lower limit for Liouville angles, and (11.4) gives the upper limit at most $1/2$.

It remains to verify unboundedness, which does not follow from a subsequence alone. Fix any proposed width K . The $2K$ phases of counts $0, \dots, 2K - 1$ are distinct for an irrational angle and have some positive minimum separation s_K . Independent words of length $2K - 1$ imply $\lfloor N/(2K - 1) \rfloor s_K^2 \leq \log(1/\kappa)$ for a width- K machine. This fails for all sufficiently large N . Thus $W_{N,\varepsilon} \rightarrow \infty$. $\square$

Corollary 13.4 (An explicit fluctuating alphabet). *Let $\alpha_L = \sum_{j=1}^{\infty} 10^{-j!}$. For $q_m = 10^{m!}$ and all large m , the horizon*

$$N_m = \lfloor q_m^{m+2}/200 \rfloor$$

admits an exact machine with at most q_m labels. The same fixed two-letter alphabet also has the cubic subsequences of Theorem 13.2; hence no single logarithmic width exponent exists. Adding reflection preserves these conclusions.

Proof. The first m summands have denominator dividing q_m , and their remaining positive tail is less than $2 \cdot 10^{-(m+1)!}$. Thus $0 < \|q_m \alpha_L\|_{\mathbb{R}/\mathbb{Z}} < 2q_m^{-m}$ for large m , proving the horizon bound by (11.3). The same tail estimates preclude rationality and show the Liouville property. The general cubic theorem applies. Reflection leaves the upper polygons invariant, while the rotation-only test words give the same lower bounds. $\square$

These theorems describe different scales of one and the same experiment. A small label count on a long resonant horizon is not a constant-width simulator for all horizons, and it need not persist across the gaps between successive approximants. The exact finite optimum and the Liouville upper logarithmic limit remain separate quantitative questions. None of the proofs substitutes approximate output for the exact upper realizations.

14. FINITE RANDOM BITS AND UNIFORM SPACE

Exact real rows and finite-bit algorithms are different resources. We first give a finite perturbation statement and then remove the row table from a concrete uniform construction. Only this section uses a computational model. It does not change the exact atomic-row conclusions proved elsewhere.

Lemma 14.1 (Dyadic perturbation of a stochastic transducer). *Suppose initialization and every command row have at most s nonzero entries. They can be replaced by denominator- 2^b rows, on the same label sets, with row-TV error at most $(s - 1)2^{-b}$. Rounding the terminal binary probabilities adds at most 2^{-b} . Consequently every numerical word changes by at most $((N + 1)(s - 1) + 1)2^{-b}$ in binary total variation. Each rounded row can be sampled with exactly b independent fair bits.*

Proof. Order the s successors, round the first $s - 1$ probabilities down to multiples of 2^{-b} , and give the remaining mass to the last successor. The total removed mass is at most $(s - 1)2^{-b}$, which is exactly the row total variation. Couple the initializations and then the command updates, or telescope products using TV contraction, to sum their worst-row errors. A terminal Bernoulli parameter can be rounded down with error at most 2^{-b} . A uniform integer in $\{0, \dots, 2^b - 1\}$ selects one of the cumulative dyadic intervals. This uses a fixed number of bits, with no retained random tape or rejection-time side channel. $\square$

For a table implementation the storage is $O(|\mathcal{A}|Ks(\log K + b))$ bits for common rows of width K , in addition to initialization and decoder data. Cut-dependent rows can require a factor N more storage. Lemma 14.1 is the elementary stochastic-rounding principle; it is not claimed as a new finite-bit simulation method. The next construction uses no such table.

Definition 14.2 (Uniform command-stream model). A fixed finite program receives N in binary, one of the four planar seeds, exactly N one-way command symbols, and one final coordinate query. It has a work tape and fresh independent fair bits. The command input cannot be reread. All counters, arithmetic registers, stored parameters, random words and output storage count toward work space. An external source waits while a command is processed; waiting time is not an input or a persistent storage channel. The finite control is fixed independently of N . The program has no horizon-dependent advice or transition table.

An angle has an admissible digit procedure if a fixed program computes a dyadic approximation within 2^{-p} in polynomial time in p and $O(p)$ work bits. The same fixed r is used throughout. This explicit hypothesis excludes a free oracle for an arbitrary noncomputable real. Algebraic angles, in particular the examples below, meet it by rational bisection with fixed-degree integer arithmetic.

Theorem 14.3 (A table-free finite-bit compiler). *Fix r angles with admissible digit procedures, rational $0 < \rho \leq 1/10$, and rational $0 < \varepsilon < 1$. There is a uniform program in Definition 14.2 whose every binary output row has TV error at most ε from the rotation experiment. It uses*

$$(14.1) \quad O_{r,\alpha,\rho}(\log(N+2) + \log(1/\varepsilon)) \text{ work bits}, \quad (N+2)b \text{ fair bits},$$

where $b = \lceil \log_2(16(N+2)/\varepsilon) \rceil$. Its total running time is polynomial in $N + \log(1/\varepsilon)$ for fixed digit procedures. Its clean stochastic register has $q+1$ labels, including one central label, with $q = O(\sqrt{N})$.

If a denominator $q_0 \geq 4$ satisfies $N\eta_\alpha(q_0) \leq q_0^2/2000$, the search may choose $q \leq q_0$. Consequently the clean-register upper exponents and logarithmic bounds of Theorem 9.1 hold for this compiler as well, with changed constants. These clean labels do not include its scratch space or deterministic parameter registers; the full work bound is (14.1).

Proof. We give the parameter search, real reference rows, dyadic rounding, and finite-space evaluation separately.

Parameter search. Put $p = \lceil \log_2(16000(N+2)/\varepsilon) \rceil$ and obtain dyadic $\tilde{\alpha}_i$ with error at most 2^{-p} . For $q = 4, 5, \dots$ compute, by integer arithmetic,

$$e(q) = \max_i \|q\tilde{\alpha}_i\|_{\mathbb{R}/\mathbb{Z}} + q2^{-p}.$$

Stop at the first q for which $Ne(q) \leq q^2/1000$. The test is rational and certifies both $\eta_\alpha(q)$ and $\eta_{\tilde{\alpha}}(q)$ by the Lipschitz property of distance to $\mathbb{Z}$. It succeeds by $q = \max(4, \lceil \sqrt{2000(N+2)} \rceil)$: $\|q\tilde{\alpha}_i\| \leq 1/2$ and the added term is smaller than one at this bound. If q_0 satisfies the stronger displayed condition, then $e(q_0) \leq \eta_\alpha(q_0) + 2q_02^{-p}$ and $2Nq_02^{-p} \leq q_0^2/2000$. Hence that candidate succeeds. No exact comparison of an irrational angle with a rational sector boundary is performed.

Reference rows with slack. Use the regular q -gon $v_j = (\cos(2\pi j/q), \sin(2\pi j/q))$ and one additional label $*$ representing zero. Set $\Lambda = 1 + 1/(4N)$. For $N \geq 1$ the dilation estimate (11.5) and the search condition give

$$\log \Lambda \geq 1/(5N) > 4\pi^2\eta_{\tilde{\alpha}}(q)/q^2.$$

Thus $\Lambda^{-1}R_{2\pi\tilde{\alpha}_i}v_j$ lies strictly inside the polygon. The case $N = 0$ is handled by direct rounding of the seed response and a fixed amount of space. Write $q\tilde{\alpha}_i = n_i + t_i$, with integer n_i and $0 \leq t_i < 1$, and put $h = 2\pi/q$. The command row has successors $j + n_i$, $j + n_i + 1$ modulo q , and $*$, with respective weights

$$(14.2) \quad A_i = \frac{\sin((1-t_i)h)}{\Lambda \sin h}, \quad B_i = \frac{\sin(t_i h)}{\Lambda \sin h}, \quad 1 - A_i - B_i.$$

They are nonnegative by the dilation bound, and their vector mean is the required contracted rotation. The row at $*$ stays there. Initialize with vector mean $x/2$ using the same chord formula with radial factor $1/2$ and the exact seed angle $0, 1/4, 1/2, 3/4$. Decode with mean $2\rho\Lambda^N \langle e_j, v_s \rangle$,

or zero at $*$. Since $\Lambda^N \leq e^{1/4}$, all decoder means are strictly inside $[-1, 1]$. The usual mean induction realizes every word for $\tilde{\alpha}$ exactly. These are reference real rows only.

Certified finite rows. Compute outward intervals for the first two weights in each row, each of width at most 2^{-b} . Clip their lower endpoints at zero and round down to denominator 2^b ; assign the residue to $*$. Relative to the reference row, at most $4 \cdot 2^{-b}$ mass is moved. Evaluate and round each terminal Bernoulli parameter within $2 \cdot 2^{-b}$. TV contraction therefore bounds initialization, N updates and final decoding by $(4N + 6)2^{-b}$. Replacing the true angles by their dyadic approximants changes a product rotation by at most $2\pi N 2^{-p}$ in operator norm, and changes a target binary row by at most $\pi \rho N 2^{-p}$. The sum is less than ε with the chosen p, b . Use exactly b fair bits at each draw, including dummy bits when the central row is deterministic. The random-bit count is $(N + 2)b$.

Work and uniform evaluation. Only q , the current vertex or central flag, the horizon and command counter, the r dyadic angles, and $O(r)$ offsets and dyadic weights are retained. The latter specify circulant rows, not a q -row table. The denominator search uses integers of $O(p + \log q)$ bits. Elementary sine and cosine on bounded intervals can be evaluated by fixed-point Taylor recurrences with outward rounding; π can be evaluated by Machin's arctangent identity. There are $O(b + \log q)$ terms and only a fixed number of live accumulators. Since $\sin(2\pi/q) \geq 4/q$ for $q \geq 4$, division loses only $O(\log q)$ bits. Repeated squaring evaluates Λ^N with $O(\log N)$ stages, all intermediate true values between 1 and $e^{1/4}$. An additional $O(\log(N + 2) + \log(b + 2))$ guard bits bounds accumulated rounding errors. Interval precision can instead be doubled until the stated widths are certified; these error estimates bound the final precision by the same order. No integer of unbounded exact denominator is stored. Every live register has $O(b + p + \log N)$ bits, and the number of registers is fixed for fixed r . This proves the work bound and a polynomial time bound, including parameter search. Finally the finite arithmetic upper proofs remain valid with $1/100$ replaced by $1/2000$, since only fixed constants change. Their candidate denominators therefore bound the search. $\square$

The proof describes a fixed-point implementation, not the numerical behavior of a floating-point library. The accompanying integer-interval reference code certifies each rounded row and uses fresh bounded-length random integers; its Python object overhead is not identified with the Turing work-space constant.

Corollary 14.4 (Sharp work-space order for explicit alphabets). *Let $\alpha_i = 2^{i/(r+1)}$, $1 \leq i \leq r$, and fix rational $0 < \varepsilon < \rho/(2\sqrt{2})$ and $0 < \rho \leq 1/10$. In Definition 14.2, numerical simulation requires and suffices with $\Theta(\log N)$ work bits. More precisely every fixed program using $s(N)$ work bits obeys*

$$(14.3) \quad s(N) \geq \frac{r}{2r+1} \log_2 N - O(\log \log(N+2)).$$

The compiler's clean register has $O(N^{r/(2r+1)})$ labels. This is not a claim that all its internal fair-bit states retain that label exponent or that the leading coefficient of work space is optimal.

Proof. Put $\theta = 2^{1/(r+1)}$. For a nonzero integer vector m with $H = \|m\|_1 \geq 1$, choose the integer p nearest to $-\sum_i m_i \theta^i$. The algebraic integer $P(\theta) = p + \sum_i m_i \theta^i$ is nonzero: the minimal polynomial $t^{r+1} - 2$ is Eisenstein and has degree $r+1$. Each of its other r conjugates has modulus at most $(9/2)H$, since $|p| \leq 2H + 1/2$ and $|\theta^i| < 2$. Its nonzero integer norm gives

$$\|m \cdot \alpha\|_{\mathbb{R}/\mathbb{Z}} \geq (2/9)^r H^{-r}.$$

This classical algebraic example is credited in [4]; it is not new arithmetic. The inherited badly approximable theorem, Corollary 11.4, now supplies the all-hidden lower bound and the matching candidate denominators for the compiler. Bisection for θ and fixed-degree powers meet the digit

hypothesis. A fixed program with $s(N)$ work bits has at most $C(s(N) + 1)^C 2^{s(N)}$ command-boundary configurations. Its private randomness since the last boundary is integrated into the next stochastic row, so it induces a legal atomic realization of at most that width. Apply the lower bound $W_{N,\varepsilon} \geq cN^{r/(2r+1)}$ and take logarithms. The polynomial factor accounts for the fixed number of work-tape head positions; it contributes $O(\log \log N)$ unless the displayed lower bound is already automatic. Theorem 14.3 supplies the upper work bound. $\square$

15. ARITHMETIC, QUANTIZATION AND NUMERICAL SEMANTICS

Kronecker quasi-uniformity. Dick–Goda–Larcher–Pillichshammer–Suzuki [4], Theorem 3.9, characterize quasi-uniformity of the multidimensional sequence $n\alpha \pmod{\mathbb{Z}^r}$ by the simultaneous bound $\eta_\alpha(q) \geq cq^{-1/r}$. Their Lemma 3.11 bounds ratios of successive best simultaneous denominators; Section 3.3 also contains the vector $(2^{1/(r+1)}, \dots, 2^{r/(r+1)})$. These are direct arithmetic antecedents, not new examples of this paper.

Our packet points instead have the form $m \cdot \alpha \pmod{1}$. The relevant separation is dual. The residue argument of Lemma 11.2, with an arbitrary nearest-integer numerator and $\eta_\alpha(q)$ in place of Q^{-1} , proves directly that a dual bound with exponent r implies

$$\eta_\alpha(q) \geq br^{-r-1}q^{-1/r}.$$

Thus the cited quasi-uniformity theorem applies to the uniform badly approximable class used in the predecessor. Its denominator control supplies the arithmetic scales for a polygon upper bound. It supplies neither a compatible stochastic transition table nor a lower bound against discarded hidden history. The logarithmic separation of Lemma 12.1 is weaker than a uniform badly approximable inequality; it is proved by a summable tail estimate, not asserted to imply quasi-uniformity.

What is dynamic in the paired profiles. The geometry of nearest-center distortion and convex hull approximation is classical [5, 11]. A static k -center factorization does not specify how its encoder is calculated without retaining discarded inputs. Conversely, a convex codebook does not force arbitrary hidden states to possess coordinates in that codebook. The two implications in Section 3 avoid both identifications: the converse conditions on the actual state, and the upper theorem provides the same legal rows throughout the run. The packing–dilation balance in Corollary 3.7 is a conditional theorem on independently specified finite profiles, not a claimed equality of static and dynamic optima.

The circle program makes one of these profiles exactly finite, whereas the all-harmonic action-gap optimization of the predecessor remains infinite-dimensional. A finite word length can be exhaustively listed; an unbounded harmonic degree cannot be certified by a finite truncation without a tail theorem. No general action-gap algorithm follows from Proposition 3.1.

Branching programs and probability preservation. At fixed horizon our machine is a probabilistic ordered read-once layered program with arbitrary stochastic rows and bounded real output columns. We distinguish width from total size, uniform description and random-bit complexity. Fourier growth for regular, permutation and width-three branching programs concerns different hypotheses and objectives [9, 10, 6]. Circle-walk harmonic damping and its arithmetic behavior are also established subjects [2]. Our converse can use correlated letters inside independently sampled complete packets; it is not a theorem about average error under fair independent letters.

Preserving a strict cutpoint sign is weaker than preserving a numerical probability. The predecessor’s constant-width sign construction contracts the output amplitude with length and hence does not contradict the present width growth. Here signal and tolerance are fixed, one query is selected after the word, and neither joint answers nor anytime correctness is required. Exact upper rows in this article are atomic real kernels, not finite-fair-bit algorithms. A row with three successors can still have costly or irrational coefficients.

Conclusions. The finite optimized word profile and the compatible enclosure profile retain quantitative information which qualitative gap data discard. They yield a minimal-type exponent and an almost-everywhere logarithmic estimate, and they exhibit fluctuating numerical width for every Liouville single angle. The reflected versions show that these conclusions survive a particular noncommuting finite extension, without asserting a classification of arbitrary noncommuting nongapped alphabets.

Exact finite widths, the Liouville upper logarithmic limit, optimal enclosure shapes and a general comparison between the two profiles remain distinct quantitative problems. The results here do not depend on a new external spectral-gap computation. The earlier expansion, calibration and representation theorems retain their original hypotheses in the unchanged supporting article.

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